

LECTURE #6
FORMAL SCATTERING THEORY

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle$$

$$|\phi_{out}\rangle = \hat{S} |\phi_{in}\rangle$$

$$|\psi(0)\rangle = |\psi\rangle$$

FOR BREVITY

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{out}\rangle$$

$$|\psi(0)\rangle = e^{+\frac{i}{\hbar} \hat{H} t} e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle \quad t \rightarrow -\infty$$

$$= |\phi_{in}\rangle - \int_t^0 dt' \left(e^{+\frac{i}{\hbar} \hat{H} t'} e^{-\frac{i}{\hbar} \hat{H}_0 t'} \right) |\phi_{in}\rangle \quad t \rightarrow -\infty = \frac{d}{dt'}$$

$$|\phi_{in}\rangle - \left(e^{+\frac{i}{\hbar} \hat{H} t'} e^{-\frac{i}{\hbar} \hat{H}_0 t'} \right) \Big|_t^0 |\phi_{in}\rangle$$

$$|\psi\rangle = |\phi_{in}\rangle - \int_{-\infty}^0 dt' \left(e^{+\frac{i}{\hbar} \hat{H} t'} e^{-\frac{i}{\hbar} \hat{H}_0 t'} \right) |\phi_{in}\rangle$$

LECTURE #6

FORMAL SCATTERING THEORY

$$\frac{d}{dt} e^{\underline{M}t} = \underline{M} e^{\underline{M}t} = e^{\underline{M}t} \underline{M}$$

$$\begin{aligned} \frac{d}{dt} \left(e^{+\frac{i}{\hbar} \hat{H} t} e^{-\frac{i}{\hbar} \hat{H}_0 t} \right) &= e^{+\frac{i}{\hbar} \hat{H} t} \hat{H} e^{-\frac{i}{\hbar} \hat{H}_0 t} \\ &+ e^{+\frac{i}{\hbar} \hat{H} t} \left(-\frac{i}{\hbar} \hat{H}_0 \right) e^{-\frac{i}{\hbar} \hat{H}_0 t} = \frac{i}{\hbar} e^{+\frac{i}{\hbar} \hat{H} t} \underbrace{\left(\hat{H} - \hat{H}_0 \right)}_{\hat{V}} e^{-\frac{i}{\hbar} \hat{H}_0 t} \end{aligned}$$

$$\hat{H} = \hat{H}_0 + \hat{V}$$

$$\hat{H}_0 = \frac{\hat{p}^2}{2m}$$

$$\hat{V} = V(\hat{x})$$

$$|\psi\rangle = |\phi_{in}\rangle - \frac{i}{\hbar} \int_{-\infty}^0 dt e^{+\frac{i}{\hbar} \hat{H} t} \hat{V} e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle$$

$$|\psi\rangle = |\phi_{in}\rangle - \int_{-\infty}^0 dt \left(e^{+\frac{i}{\hbar} \hat{H} t} e^{-\frac{i}{\hbar} \hat{H}_0 t} \right) |\phi_{in}\rangle$$

||
|ψ(0)⟩

$$|\phi_{in}\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^{FREE}\rangle$$

$$\phi_{in}(x) = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x}$$

$$\Phi_{in}(p) = \langle \psi_p^{FREE} | \phi_{in} \rangle$$

$\hat{H}_0$ EIGENPROBLEM

$$\hat{H}_0 \psi_p^{\text{FREE}}(x) = E_p \psi_p^{\text{FREE}}(x)$$

$$\psi_p^{\text{FREE}}(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x}$$

$$E_p = \frac{p^2}{2m}$$

$$\hat{H}_0 |\psi_p^{\text{FREE}}\rangle = E_p |\psi_p^{\text{FREE}}\rangle$$

$p \in \mathbb{R}$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\psi_p^{\text{FREE}}\rangle = e^{-\frac{i}{\hbar} E_p t} |\psi_p^{\text{FREE}}\rangle$$

$$|\psi\rangle = |\phi_{\text{in}}\rangle - \frac{i}{\hbar} \int_{-\ell}^0 dt e^{+\frac{i}{\hbar} \hat{H} t} \hat{V} e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{\text{in}}\rangle$$

~~$$|\psi\rangle = \int_{-\ell}^{+\ell} dp \Phi_{\text{in}}(p) |\psi_p^{\text{FREE}}\rangle - \frac{i}{\hbar} \int_{-\ell}^0 dt e^{+\frac{i}{\hbar} \hat{H} t} \hat{V} e^{-\frac{i}{\hbar} \hat{H}_0 t} \int_{-\ell}^{+\ell} dp \Phi_{\text{in}}(p) |\psi_p^{\text{FREE}}\rangle$$~~

$$|\psi\rangle = \int_{-\ell}^{+\ell} dp \Phi_{\text{in}}(p) |\psi_p^{\text{FREE}}\rangle - \frac{i}{\hbar} \int_{-\ell}^{+\ell} dp \Phi_{\text{in}}(p) \int_{-\ell}^0 dt e^{+\frac{i}{\hbar} \hat{H} t} \hat{V} \underbrace{e^{-\frac{i}{\hbar} \hat{H}_0 t}}_{e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}} |\psi_p^{\text{FREE}}\rangle$$

$$e^{+\frac{i}{\hbar} \epsilon t}$$

$$\epsilon > 0$$

$$\epsilon \rightarrow 0$$

$$\lim_{\epsilon \rightarrow 0_+} \int_{-\ell}^0 e^{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - 5 + i\epsilon \right) t} dt =$$

$$= \frac{e^{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - 5 + i\epsilon \right) t}}{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - 5 + i\epsilon \right)} \bigg|_{t=-\ell}^{t=0}$$

$$= \frac{1}{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - 5 + i\epsilon \right)}$$

$$|\psi\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^{FREE}\rangle - \frac{i}{\hbar} \int_{-\infty}^{+\infty} dp \Phi_{in}(p) \left(\int_{-\infty}^0 dt e^{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - \hat{H} + i\varepsilon \right) t} \right) |\psi_p^{FREE}\rangle$$

$\varepsilon \rightarrow 0_+$

$$\int_{-\infty}^0 e^{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - \hat{H} + i\varepsilon \right) t} dt = \frac{e^{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - \hat{H} + i\varepsilon \right) t}}{\left(-\frac{i}{\hbar} \right) \left(\frac{p^2}{2m} - \hat{H} + i\varepsilon \right)} \Big|_{-\infty}^0 = \frac{1}{-\frac{i}{\hbar} \left(\frac{p^2}{2m} - \hat{H} + i\varepsilon \right)} \quad \hat{V} = V(\hat{x})$$

$\varepsilon \rightarrow 0_+$

$$\hat{A} \dots \frac{1}{\hat{A}} = \hat{A}^{-1}$$

$$\frac{1}{\frac{p^2}{2m} - \hat{H} + i\varepsilon} = \left(\frac{\hat{p}^2}{2m} - \hat{H} + i\varepsilon \right)^{-1} V(\hat{x}) e^{-\varepsilon|t|}$$

~~$$|\psi\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^{FREE}\rangle - \frac{i}{\hbar} \int_{-\infty}^{+\infty} dp \Phi_{in}(p) \int_{-\infty}^0 dt e^{-\frac{i}{\hbar} \hat{H} t} \left(\frac{e^{-\frac{i}{\hbar} \hat{H} t}}{e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}} \right) |\psi_p^{FREE}\rangle$$~~

$$\hat{H} |\psi_p\rangle = \frac{p^2}{2m} |\psi_p\rangle$$

$$\lim_{t \rightarrow -\infty} e^{+\varepsilon t} = 0 \quad (\varepsilon > 0)$$

$$f(\hat{H}) = \int_{-\infty}^{+\infty} dp' |\psi_{p'}\rangle \left(\frac{p'^2}{2m} \right) \langle \psi_{p'}|$$

$$\langle x | \frac{1}{\frac{p^2}{2m} - \hat{H} + i\varepsilon} | x' \rangle = \int_{-\infty}^{+\infty} dp' \frac{\psi_{p'}(x) \psi_{p'}^*(x')}{\frac{p^2}{2m} - \frac{p'^2}{2m} + i\varepsilon}$$

$$|\psi\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^{FREE}\rangle + \int_{-\infty}^{+\infty} dp \Phi_{in}(p) \left(\frac{1}{\frac{p^2}{2m} - \hat{H} + i\varepsilon} \right) |\psi_p^{FREE}\rangle$$

$$|\psi\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) \left(\hat{1} + \frac{1}{\frac{p^2}{2m} - \hat{H} + i\varepsilon} \hat{V} \right) |\psi_p^{FREE}\rangle \quad \varepsilon \rightarrow 0_+$$

DEFINE NOW

~~LIPP MANN - SCHWINGER EQUATION~~

$$|\psi_p^\pm\rangle = |\psi_p^{FREE}\rangle + \frac{1}{\frac{p^2}{2m} - \hat{H} \pm i\varepsilon} \hat{V} |\psi_p^{FREE}\rangle$$

$$|\phi_{in}\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^{FREE}\rangle$$

$$|\psi\rangle = \int_{-\infty}^{+\infty} dp \Phi_{in}(p) |\psi_p^+\rangle$$

$$= \int_{-\infty}^{+\infty} dp \Phi_{out}(p) |\psi_p^-\rangle$$

$$|\phi_{out}\rangle = \int_{-\infty}^{+\infty} dp \Phi_{out}(p) |\psi_p^{FREE}\rangle$$

$$\frac{1}{\frac{p^2}{2m} - \hat{H} + i\varepsilon} = \left(\frac{p^2}{2m} - \hat{H} + i\varepsilon \right)^{-1} \quad \text{RETARDED GREEN OPERATOR}$$

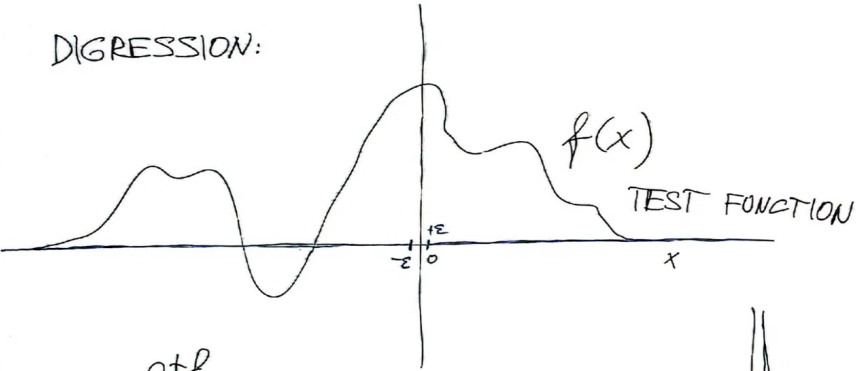
$$\frac{1}{\frac{p^2}{2m} - \hat{H} - i\varepsilon} = \left(\frac{p^2}{2m} - \hat{H} - i\varepsilon \right)^{-1} \quad \text{ADVANCED GREEN OPERATOR}$$

POSITION REPRESENTATIONS

$$\langle x | \frac{1}{E - \hat{H} \pm i\varepsilon} | x' \rangle = G_E^{(\pm)}(x, x')$$

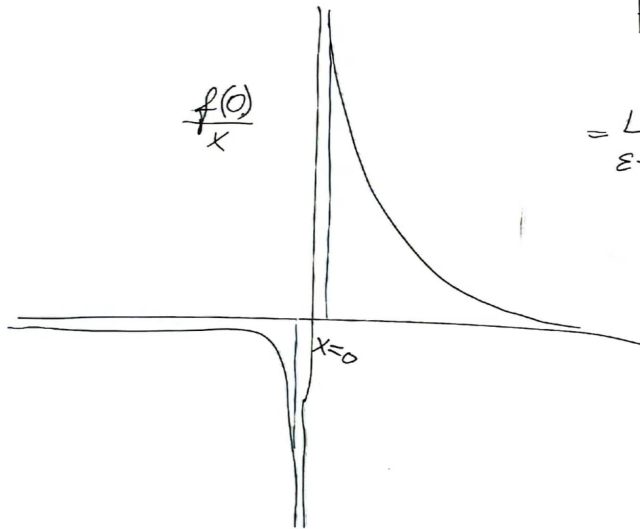
RETARDED / ADVANCED GREEN FUNCTIONS

DIGRESSION:



$$\int_{-\infty}^{+\infty} \frac{f(x)}{x} dx$$

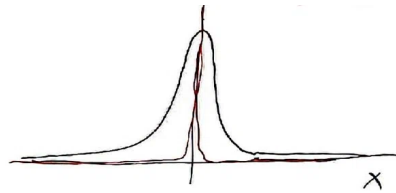
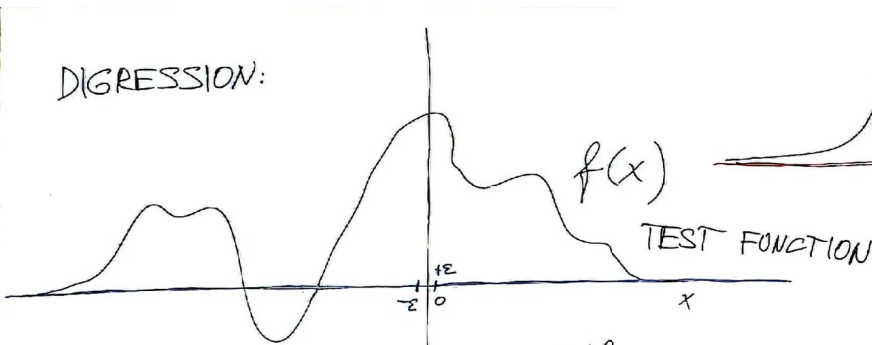
$$\frac{f(0)}{x}$$



CAUCHY PRINCIPAL PART

$$P \int_{-\infty}^{+\infty} dx \frac{f(x)}{x} = \lim_{\varepsilon \rightarrow 0^+} \left(\int_{-\infty}^{-\varepsilon} dx \frac{f(x)}{x} + \int_{\varepsilon}^{+\infty} dx \frac{f(x)}{x} \right)$$

DIGRESSION:



$$\frac{1}{x \pm i\varepsilon} = P \frac{1}{x} \mp i\pi \delta(x)$$

CAUCHY PRINCIPAL PART

$$P \int_{-\infty}^{+\infty} dx \frac{f(x)}{x} = \lim_{\varepsilon \rightarrow 0^+} \left(\int_{-\infty}^{-\varepsilon} dx \frac{f(x)}{x} + \int_{\varepsilon}^{+\infty} dx \frac{f(x)}{x} \right)$$

$$\int_{-\infty}^{+\infty} dx \frac{f(x)}{x \pm i\varepsilon}$$

$$\frac{1}{(x \pm i\varepsilon)} \frac{(x \mp i\varepsilon)}{(x \mp i\varepsilon)} = \frac{x \mp i\varepsilon}{x^2 + \varepsilon^2}$$

$$= \frac{x}{x^2 + \varepsilon^2} \mp i\pi \left(\frac{\varepsilon}{x^2 + \varepsilon^2} \frac{1}{\pi} \right)$$

$$\int_{-\infty}^{+\infty} dx \frac{f(x) x}{x^2 + \varepsilon^2}$$

$$\mp i\pi \int_{-\infty}^{+\infty} dx f(x) \frac{1}{\pi} \frac{\varepsilon}{x^2 + \varepsilon^2}$$

